INCIDENT RESPONSE REPORT

Incident Time: 2300 hrs

Location: FCTS

1. **Initial Event**. Network Reconnaissance (Nmap Scan)

    1.1. The **Nmap SYN** scan indicates the attacker was enumerating open ports and services.

    1.2. This is typically the first stage of the cyber kill chain (reconnaissance phase).

    1.3. **Conclusion**. The attack began with port scanning to identify vulnerable services

2. **Evidence**. Telnet Traffic Capture

    2.1. Telnet transmits credentials in plaintext

    2.2. Packet analysis (via Wireshark logs) shows:

        2.2.1. username: usersushil

        2.2.2. password: sushil

    2.3. Supporting Indicators:

        2.3.1. Suricata alert: *"Telnet Traffic Detected"* sid:100003

        2.3.2. Visible credentials in TCP stream.

        2.3.3. **Conclusion**. Credential theft is confirmed through plaintext Telnet session capture

3. **Attacker Identification**: Unknown External Host

    3.1. Source IP observed in following: -
        3.1.1. Nmap scan logs
        3.1.2. Telnet session initiation
        3.1.3. Connection attempts on port 4444

    3.2. Indicators.
        3.2.1. Repeated connections from same IP
        3.2.2. Port scanning behavior

3.2.3. Unauthorized service interaction

3.3. **Conclusion**. The attacker is the external IP performing scanning + unauthorized access attempts

4. **Multiple Control Failures**.

4.1. Lack of Encryption. Telnet and FTP used → plaintext exposure

4.2. No Network Segmentation. Attacker could reach multiple sectors (HQ, Operations, Logistics)

4.3. Weak IDS/IPS Response. Suricata detected activity but did not block it

4.4. No Egress Filtering. Laptop connected to suspicious port 4444 (possible reverse shell)

4.5. **Conclusion**. Failure of preventive controls (encryption, segmentation, firewall rules)

5. **Immediate Containment Actions**

5.1. **Isolate Affected Host**. Disconnect laptop from network

5.2. **Block Attacker IP**. Apply firewall rule: deny ip <attacker-ip> any

5.3. **Disable Vulnerable Services**. Stop Telnet and FTP

5.4. **Kill Suspicious Process**. Identify process on port 4444 and terminate immediately.

5.5. Enable Monitoring. Start packet capture and monitor logs in real-time

6. **Long-term remediation plan**

6.1. **Security Improvements**

6.1.1. Replace insecure protocols (Telnet, FTP)

6.1.2. Enforce strong authentication (MFA)

6.2. **Network Hardening**. Implement VLAN segmentation and Zero Trust model

6.3. **Detection & Prevention**.

6.3.1 Upgrade IDS → IPS mode
6.3.2. Integrate logs with SIEM (e.g., Wazuh)

6.4. **Endpoint Security**.

6.4.1. Install EDR solutions

6.4.2. Monitor unusual ports (e.g., 4444)

6.5. **Training**. User awareness on credential security

7. **Protocols to be replaced**.

Ser No	Insecure	Replace With	Reason
1.	Telnet	SSH	Encrypts credentials
2.	FTP	SFTP / FTPS	Secure file transfer
3.	HTTP	HTTPS	Protects data in transit

8. **Improve monitoring posture**.

8.1. **Centralized Logging**. Use SIEM (Wazuh)

8.2. **Advanced Detection Rules**. Detect following: -

8.2.1. Brute force attempts

8.2.2. Port scans

8.2.3. Data exfiltration

8.3. **Network Visibility**. Deploy following measures: -

8.3.1. IDS/IPS sensors

8.3.2. NetFlow monitoring

8.4. **Alert Prioritization**.High severity alerts.

8.4.1. Port scans

8.4.2. Credential exposure

8.4.3. Reverse shell (port 4444)

8.5. **Automation**.

8.5.1. Auto-block attacker IPs

8.5.2. Trigger incident response playbooks